Nothing Here

Fast Quantum Algorithms

or

How we learned to put our pants on two legs at a time.



Dave Bacon

Institute for Quantum Information

California Institute of Technology

?

A sudden bold and unexpected question doth many times surprise a man and lay him open.

A prudent question is one-half of wisdom.



Sir Francis Bacon
(1561-1628)

Iway amway Akespeareshay!



William Shakespeare
(1568-1623)

"small Latin, less Greek" ?

WarNING

This Talk Under Constant Acceleration

DB and CBSSS assume no responsibility for injuries sustained while zoning out.

Quantum Computers Can Do Amazing Things!

THIS TALK

Understanding what makes quantum evolution *different*.

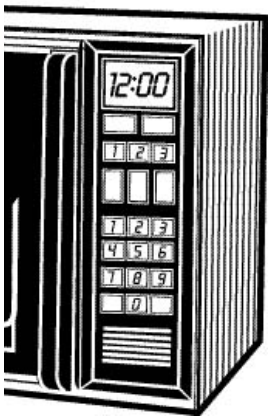
How quantum evolution can used to do something cool.

How quantum evolution can be used to exponentially speed up an oracle problem over classical deterministic algorithms.

How quantum evolution can be used to exponentially speed up an oracle problem over classical probabilistic algorithms.

Randomizing Microwave





Digital Coffee
(Not Java!)

Mystery Markov Microwave







Scalding Hot

H

Freezing Cold

C

Marko

The true method of knowledge is experiment. - William Blake 1788

V

- Run Experiments To Understand MMM Machine

If you put in C, 70% of the time you get H out
and 30% of the time you get C out
If you put in H, 80% of the time you get H out
and 20% of the time you get C out

- A nice little formalism

H $\longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

C $\longrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$M_{mm} = \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}$$

$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}$$

$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}$$

columns sum to 1

$0 \leq$ matrix entry ≤ 1

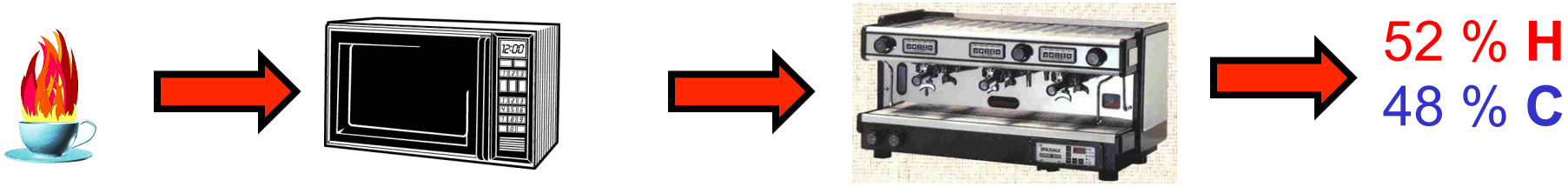
Markov Chains



$$\begin{pmatrix}1\\0\end{pmatrix} \quad \begin{pmatrix}0.8 & 0.7\\0.2 & 0.3\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}0.8\\0.2\end{pmatrix} \quad \begin{pmatrix}0.8 & 0.7\\0.2 & 0.3\end{pmatrix}\begin{pmatrix}0.8\\0.2\end{pmatrix}=\begin{pmatrix}0.78\\0.22\end{pmatrix}$$

or

$$\begin{pmatrix}0.8 & 0.7\\0.2 & 0.3\end{pmatrix}\begin{pmatrix}0.8 & 0.7\\0.2 & 0.3\end{pmatrix}=\begin{pmatrix}0.78 & 0.77\\0.22 & 0.23\end{pmatrix} \quad \begin{pmatrix}0.78 & 0.77\\0.22 & 0.23\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}0.78\\0.22\end{pmatrix}$$



$$\begin{pmatrix}1\\0\end{pmatrix} \quad \begin{pmatrix}0.8 & 0.7\\0.2 & 0.3\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}0.8\\0.2\end{pmatrix} \quad \begin{pmatrix}0.5 & 0.6\\0.5 & 0.4\end{pmatrix}\begin{pmatrix}0.8\\0.2\end{pmatrix}=\begin{pmatrix}0.52\\0.48\end{pmatrix}$$

or

$$\begin{pmatrix}0.5 & 0.6\\0.5 & 0.4\end{pmatrix}\begin{pmatrix}0.8 & 0.7\\0.2 & 0.3\end{pmatrix}=\begin{pmatrix}0.52 & 0.53\\0.48 & 0.47\end{pmatrix} \quad \begin{pmatrix}0.52 & 0.53\\0.48 & 0.47\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}0.52\\0.48\end{pmatrix}$$

Quantum Microwave

Quantum Digital Coffee



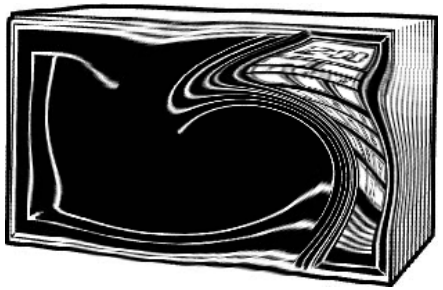
Scalding Hot

H



Freezing Cold

C



Quantum Microwave (QM)

What are the rules for the Quantum Microwave?

The Amplitude Attitude

H **C**

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$|0\rangle$ $|1\rangle$

qubit

$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

Complex numbers $(a + bi)$ ■

$\alpha|0\rangle + \beta|1\rangle$

If we measure the state then

with probability $|\alpha|^2$ we get H

with probability $|\beta|^2$ we get C

$|\alpha|^2 + |\beta|^2 = 1$

For Our Purposes

$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

Real numbers ■

$-1 \leq \alpha \leq 1$

$-1 \leq \beta \leq 1$

$\alpha|0\rangle + \beta|1\rangle$

$\alpha^2 + \beta^2 = 1$

If we measure the state then

with probability α^2 we get H

with probability β^2 we get C

Expert Note: Quantum computation with real numbers is equivalent to quantum computation with complex numbers as a consequence of $SU(N) \subseteq SO(2N)$.

Unitary



$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{\alpha+\beta}{\sqrt{2}} \\ \frac{\alpha-\beta}{\sqrt{2}} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \frac{\alpha+\beta}{\sqrt{2}} \\ \frac{\alpha-\beta}{\sqrt{2}} \end{pmatrix}$$

In general this matrix U must be unitary:

$$UU^\dagger = I$$

i.e. the rows U are orthogonal vectors.

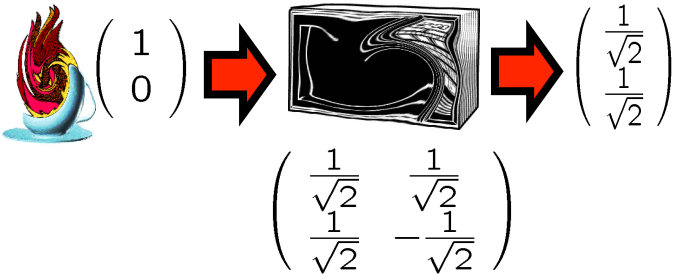
For our purposes (real numbers), this becomes

$$UU^T = I$$

Matrix entries will be between -1 and 1

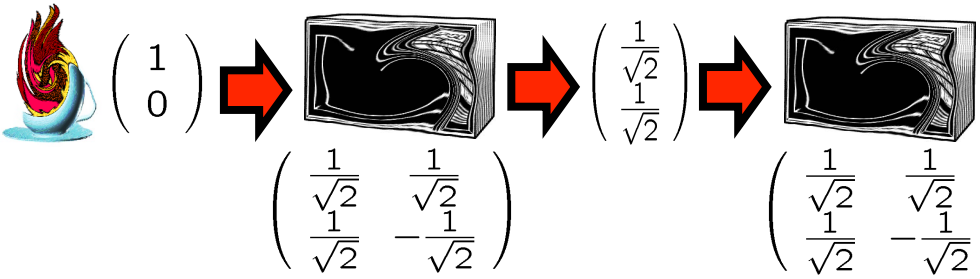
Interference





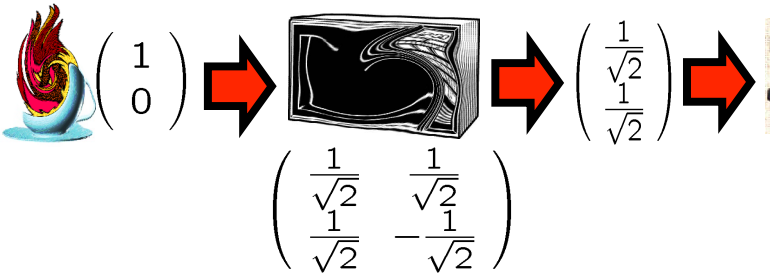
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

50 % **H**
50 % **C**



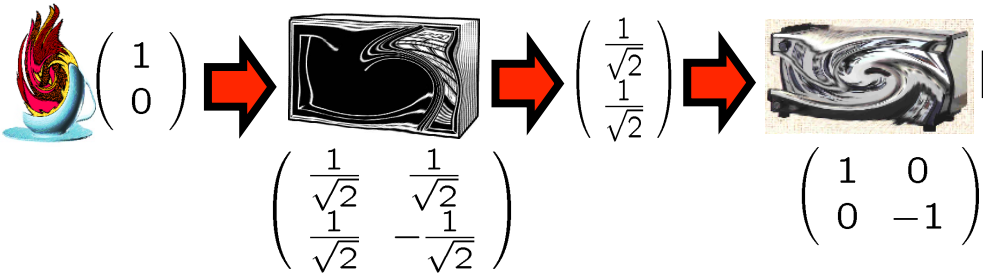
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

100 %
H
0 % **C**



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

50 % **H**
50 % **C**



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

0 % **H**
100 %
C

Deutsch's Problem



David Deutsch



Dr. Falcon

(1985)

two qubits

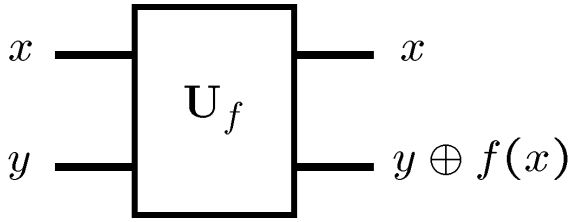
$$\alpha_0|00\rangle + \alpha_1|01\rangle + \alpha_2|10\rangle + \alpha_3|11\rangle$$

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$$



Delphi

Oracle



$x, y, f(x) \in \{0, 1\}$

Example $f(x) = x$:

$$U_f = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Four possible functions $f(x)$:

$f(x) = 0$ $f(x) = 1$ $f(x) = x$ $f(x) = \bar{x}$

Constant functions Balanced functions

Deutsch's Problem

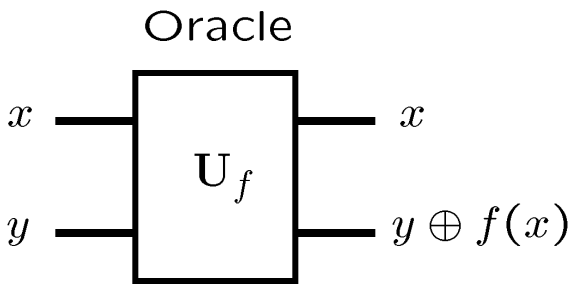
Determine whether f(x) is constant or balanced using as few queries to the oracle as possible.

Classical Deutsch

Four possible functions $f(x)$:

$f(x) = 0 \quad f(x) = 1 \quad f(x) = x \quad f(x) = \bar{x}$

Constant functions Balanced functions



$x, y, f(x) \in \{0, 1\}$

Query input of x_0 and y_0 only gives information about $f(x_0)$.

Knowing $f(x_0)$ not enough to distinguish constant from balanced.

Classically we need to query the oracle <u>two</u> times to solve Deutsch's Problem

Quantum Deutsch

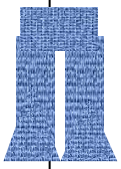


1. Query oracle with $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
2. Producing the state $\frac{1}{2}\left(|0f(0)\rangle - |0\bar{f}(0)\rangle + |1f(1)\rangle - |1\bar{f}(1)\rangle\right)$

$f(x) = 0$	$f(x) = 1$	$f(x) = x$	$f(x) = \bar{x}$
$\frac{1}{2}(\|00\rangle - \|01\rangle + \|10\rangle - \|11\rangle)$	$\frac{1}{2}(\|01\rangle - \|00\rangle + \|11\rangle - \|10\rangle)$	$\frac{1}{2}(\|00\rangle - \|01\rangle + \|11\rangle - \|10\rangle)$	$\frac{1}{2}(\|01\rangle - \|00\rangle + \|10\rangle - \|11\rangle)$
$\frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix}$
Constant functions		Balanced functions	

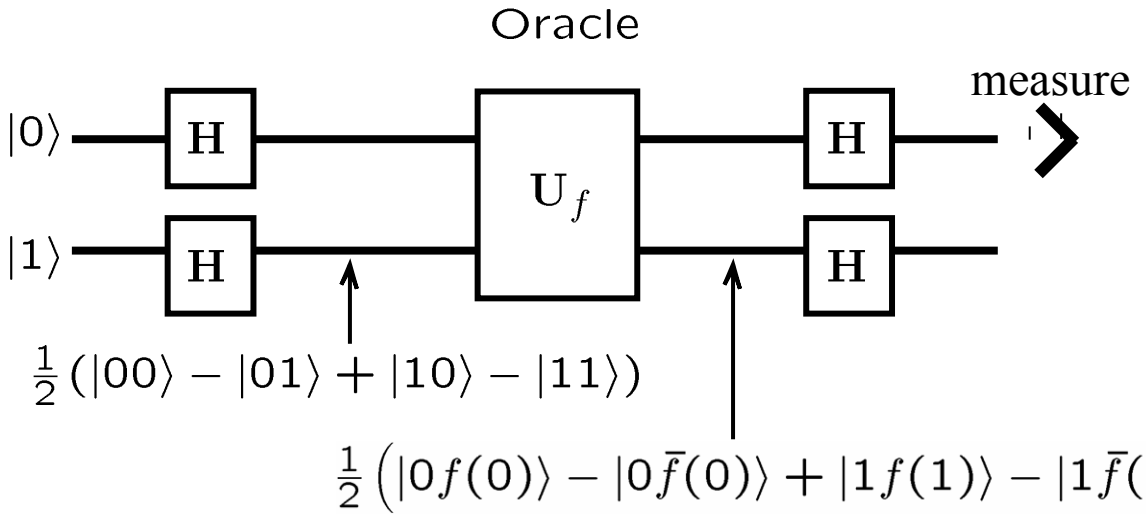
3. Apply the unitary transformation

$$\frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

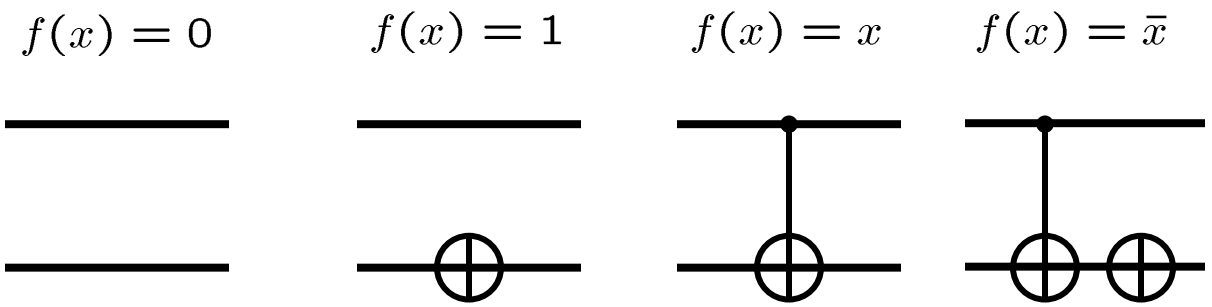


$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$
100 %	100 %	100 %	100 %

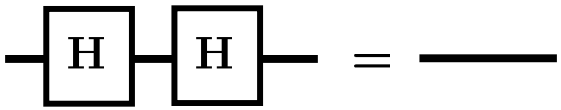
Deutsch Circuit



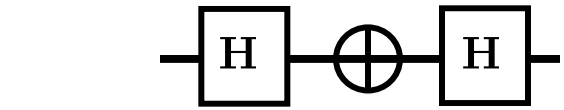
$$\mathbf{H} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \qquad \oplus = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



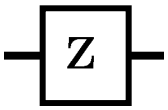
A Different View



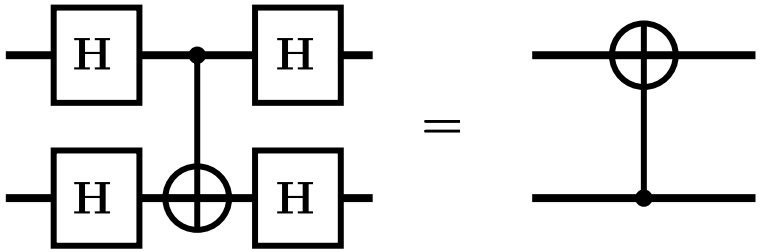
$=$



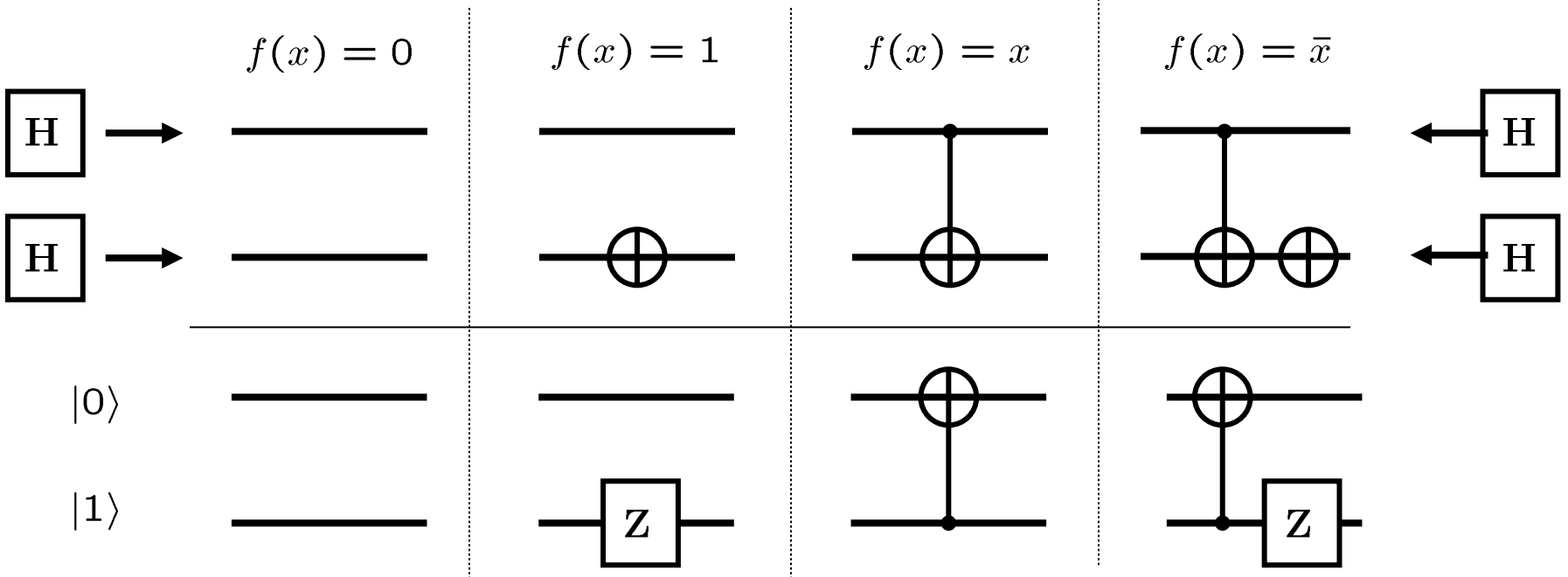
$=$



$$\left(\begin{array}{cc}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{array}\right)\left(\begin{array}{cc}0 & 1\\ 1 & 0\end{array}\right)\left(\begin{array}{cc}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{array}\right)=\left(\begin{array}{cc}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{array}\right)\left(\begin{array}{cc}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{array}\right)=\left(\begin{array}{cc}1 & 0\\ 0 & -1\end{array}\right)$$



$=$



Deutsch In Perspective



Quantum theory allows us to do in a single query what classically requires two queries.

What about problems where the computational complexity is exponentially more efficient?

Deutsch-Jozsa Problem

(1992)

Given a function $f : \{0,1\}^n \rightarrow \{0,1\}$

function maps n bit strings to a single bit

f is promised to be either

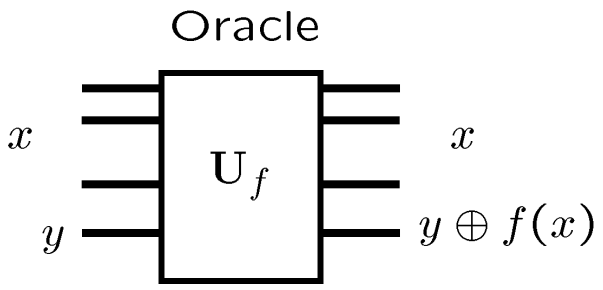
(A) f is constant.

$f(x) = b \ \forall x$

(B) f is balanced.

$f(x) = 1$ if $x \in \mathcal{S}$ otherwise $f(x) = 0$

$\mathcal{S}$ has 2^{n-1} elements.



$x \in \{0,1\}^n$

$y, f(x) \in \{0,1\}$

Deutsch-Jozsa Problem

Determine whether f(x) is constant or balanced using as few queries
to the oracle as possible.

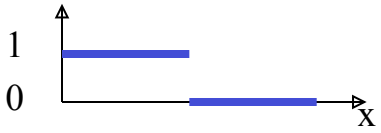
Classical DJ





(A) f is constant.

$f(x) = b \ \forall x$



(B) f is balanced.

$f(x) = 1$ if $x \in \mathcal{S}$ otherwise $f(x) = 0$

$\mathcal{S}$ has 2^{n-1} elements.

If we never want to be wrong:

Worst case we need $2^{n-1} + 1$ querries

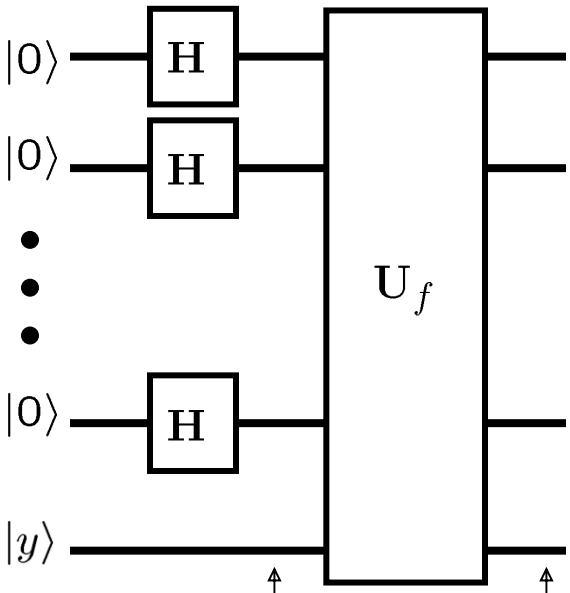
If we allow a failure probability of ϵ, then

Randomly choose k different x_k to query.

$k = O\left(log\left(\frac{1}{\epsilon}\right)\right)$ (question on website!)

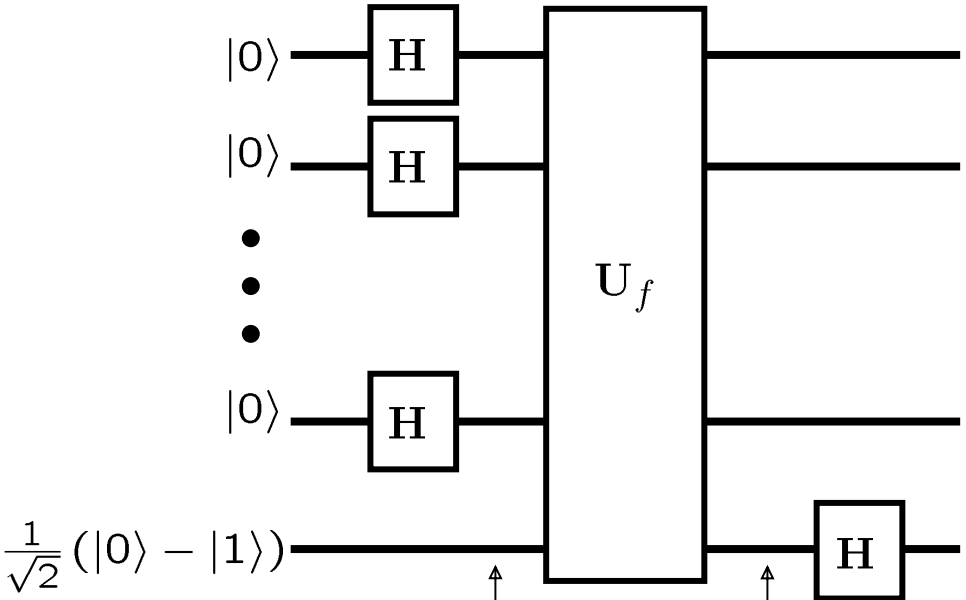
Determinstically hard, probabilistically easy.

Quantum DJ



$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|y\rangle \quad \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|y \bigoplus f(x)\rangle$$

$$|x\rangle = |x_n x_{n-1} \ldots x_1\rangle$$



$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|0\rangle - |1\rangle\right)$$

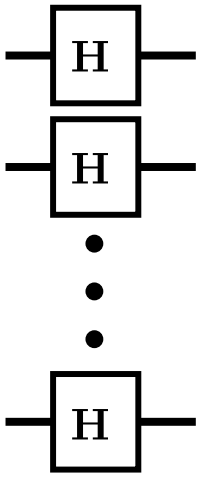
$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|f(x)\rangle - |\bar{f}(x)\rangle\right)$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle|1\rangle$$

Quantum DJ

Information about $f(x)$ is in the phase

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle |1\rangle$$



$= \mathbf{H}^{\otimes n}$

$$\mathbf{H}^{\otimes n}|x_n x_{n-1} \dots x_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_n}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_{n-1}}|1\rangle) \cdots \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_1}|1\rangle)$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x \cdot y} |y\rangle \qquad x \cdot y = x_n y_n + x_{n-1} y_{n-1} + \cdots + x_1 y_1 \mod 2$$

All matrix elements of $\mathbf{H}^{\otimes n}$ are $\pm \frac{1}{\sqrt{2^n}}$

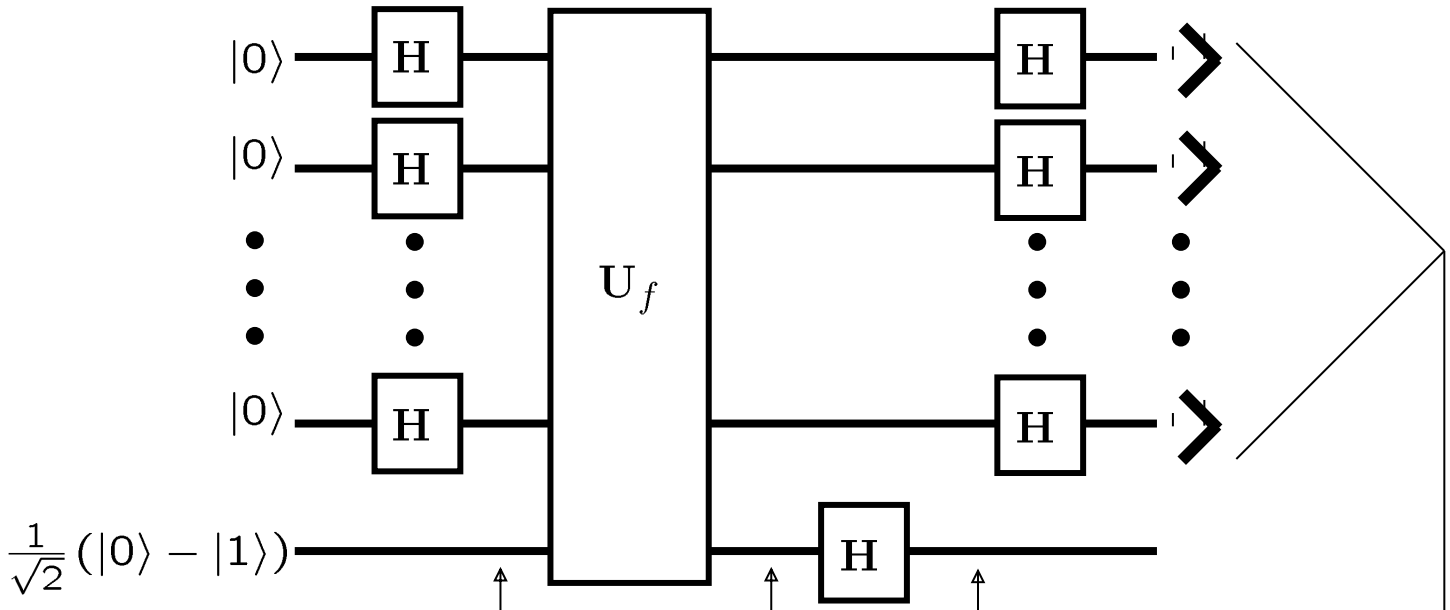
First row of $\mathbf{H}^{\otimes n}$ has all elements $+\frac{1}{\sqrt{2^n}}$

Other rows have equal number $\pm \frac{1}{\sqrt{2^n}}$ elements.

If $f(x)$ is constant, applying $\mathbf{H}^{\otimes n}$ to $\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$ always gives $|0\rangle$.

If $f(x)$ is balanced, applying $\mathbf{H}^{\otimes n}$ to $\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$ always gives a superposition without $|0\rangle$.

Full Quantum DJ



$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|0\rangle - |1\rangle\right)$$

$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|f(x)\rangle - |\bar{f}(x)\rangle\right)$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle|1\rangle$$

If all bits are 0, then f is constant.

If all bits are not 0, then f is balanced.

Solves DJ with a SINGLE query vs 2n-1+1 classical deterministic!!!!!!!!

Simon's Problem

(is that no one does what "Simon says"?) (1994)

Given a function $f : \{0,1\}^n \rightarrow \{0,1\}^n$

function maps n bit strings to a n bit strings

f is promised to satisfy either

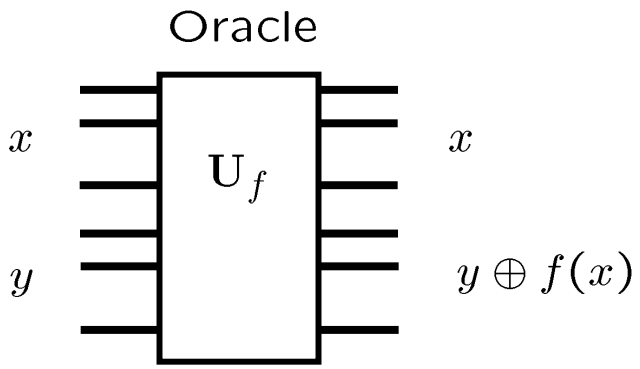
(A) f is distinct on all inputs.

$$f(x) \neq f(y) \quad \forall x \neq y$$

(B) f is distinct up to an XOR mask s

$$f(x) = f(x \oplus s)\ \forall x \qquad (s \text{ is unkown and } \neq 0)$$

$$f(x) \neq f(x \oplus t)\ \forall t \neq s$$



Simon's Problem

Determine whether f(x) has is distinct on an XOR mask or distinct
on all inputs using the fewest queries of the oracle. (Find s)

Classical Simon

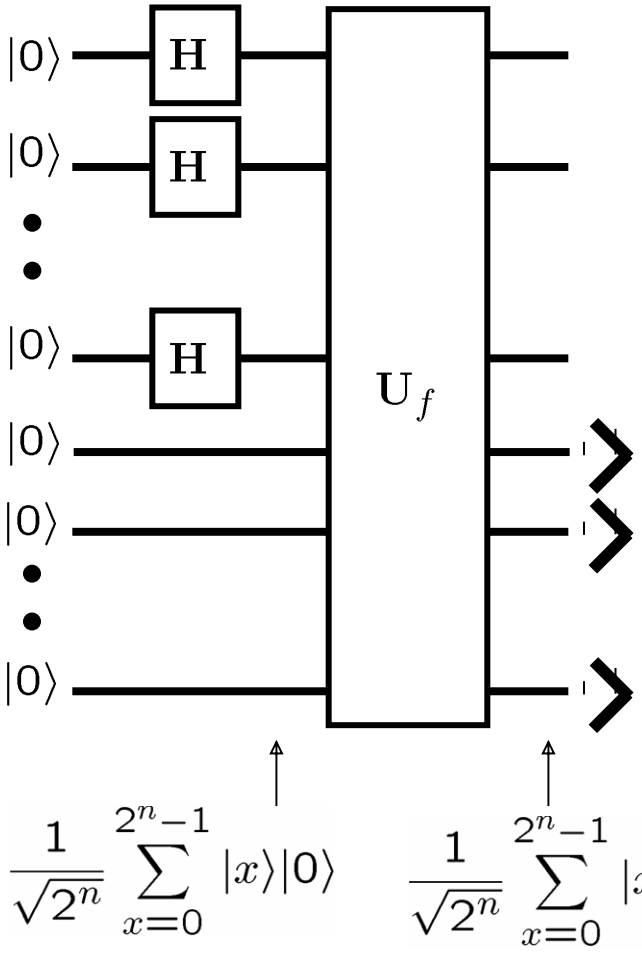


Now both determinsitcally and probabilistically an exponential number of queries is required.

Left as an exercise for those who are asleep.

Make sure to tell the sleepers that they missed a cameo appearance by my good friend Robin Williams.

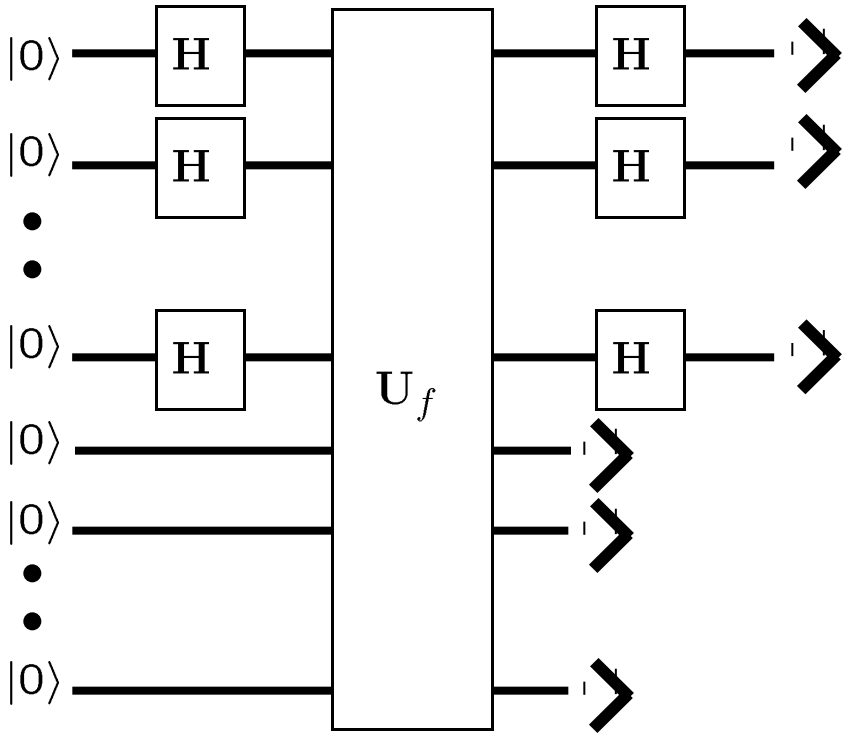
Quantum Simon



If $f(x)$ is distinct, then first register will be random $|x\rangle$.

If $f(x)$ is distinct on an XOR map s, then first register will be $\frac{1}{\sqrt{2}}(|x\rangle + |x \oplus s\rangle)$ for a random x.

Quantum Simon



Quantum Simon

$$\mathbf{H}^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x\cdot y}|y\rangle \longleftarrow \text{Equal amplitude and uniform over all } y$$

$$\mathbf{H}^{\otimes n}\frac{1}{\sqrt{2}}(|x\rangle + |x\oplus s\rangle) = \frac{1}{\sqrt{2^{n+1}}} \sum_{y=0}^{2^n-1} \left((-1)^{x\cdot y}|y\rangle + (-1)^{(x\oplus s)\cdot y}|y\rangle\right)$$

$$= \frac{1}{\sqrt{2^{n+1}}} \sum_{y=0}^{2^n-1} (-1)^{x\cdot y}(1 + (-1)^{s\cdot y})|y\rangle \quad (1)$$

Equal amplitude and uniform over all y such that $y \cdot s = 0$.

If we get $n-1$ linearly independent y's such that $y \cdot s = 0$ we can determine s.

Claim: with probability at most $\frac{1}{4}$, $n-1$ random vectors y such that $y \cdot s = 0$ are linearly independent.

An Open Question

(you could be famous!)

Given a quantum oracle which produces either

(A) $|x\rangle$

$x, l \in \{0, 2^n - 1\}$

(B) $\frac{1}{\sqrt{2}}(|x\rangle + |x + l\rangle)$ l is unkown

both for uniform random $x \in \{0, 2^n - 1\}$

Distinguishing between (A) and (B) using only a polynomial number of queries to the quantum oracle and with an efficient quantum circuit solves the

Dihydral hidden subgroup problem

which is a big step towards solving

Graph Isomorphism

Shor Type Algorithms

1) Deutsch's algorithm demonstrates task quantum computer can perform in one shot that classically takes two shots.

1992 Deutsch-Jozsa algorithm demonstrates an exponential separation between classical deterministic and quantum algorithms.

1993 Bernstein-Vazirani demonstrates a superpolynomial algorithm separation between probabilistic and quantum algorithms.

1994 Simon's algorithm demonstrates an exponential separation between probabilistic and quantum algorithms.

1994 Shor's algorithm demonstrates that quantum computers can efficiently factor numbers.

Sample Quantum Communication Complexity

B: y0y1

A: x0x1

C: z0z1

Three parties A, B, C given inputs x,y,z

Want to compute f(x,y,z) via a set protocol of communication.

Ability to "broadcast" information to other two parties. cost=# bits broadcast

SAMPLE WHERE PRESHARED ENTANGLEMENT LOWERS COST

A, B, C each given a two bit string.

guarantee: x0y0z0∈{000, 011, 101, 110}, x1y1z1 unrestricted

f(x,y,z)= x1⊕y1⊕z1⊕(x0∨y0∨z0) (⊕ is XOR, ∨ is OR)

Quantum: each party has one part of a tripartite entangled state:

$$|\psi\rangle = \tfrac{1}{2}\left(|000\rangle - |011\rangle - |101\rangle - |110\rangle\right)$$

$$|abc\rangle = |a\rangle \otimes |b\rangle \otimes |c\rangle$$

A B C

Protocol:

$$\text{Hadamard} \equiv H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

1. For each given party, if first bit (x0,y0, or z0) is 1, then apply the Hadamard gate to given part of $|\psi\rangle$

2. Next, measure the respective qubit. Denote the given parties output as a,b,c respectively.

If x0y0z0=000, then $|\psi\rangle$ unchanged, a⊕b⊕c=0

If x0y0z0=110, then $|\psi\rangle \rightarrow \frac{1}{2}\left(|010\rangle - |001\rangle + |111\rangle + |100\rangle\right)$, a⊕b⊕c=1 (etc)

a⊕b⊕c= x0∨y0∨z0

3. Parties broadcast- A: (x1⊕a) B: (y1⊕b) C: (z1⊕c)

Each party can now compute

(x1⊕a)⊕(y1⊕b)⊕(z1⊕c)= x1⊕y1⊕z1⊕(x0∨y0∨z0)

f(x,y,z) with 3 bits classical result requires: 4 bits

Buhrman, Cleve, Tapp 1997

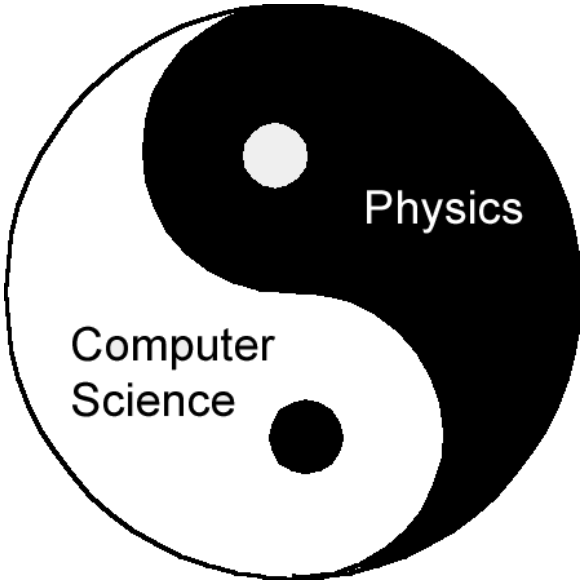
Quantum Communication Complexity

Less communication needed to compute certain functions if either (a) qubit used to communicate or (b) shared entangled quantum states are available.

How much less communication?

Exponentially less: Ran Raz “Exponential Separation of Quantum and Classical Communication Complexity”, 1998

A final word from my sponsors



Physics says to computer science, “your information carriers should be quantum mechanical” and out pops quantum computation!

What can computer science tell us about physics?!?!

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